Q: What is the primary purpose of the React Context API?
A: To share data between components in a React application
Explanation: The React Context API is used to share data between components in a React application, allowing them to access and update the same data without having to pass props down manually.

Q: What is the difference between a 'let' and a 'const' variable in JavaScript?
A: A 'let' variable can be reassigned, while a 'const' variable cannot
Explanation: In JavaScript, 'let' variables are block-scoped and can be reassigned, while 'const' variables are also block-scoped but cannot be reassigned.

Q: What is the purpose of the 'useCallback' hook in React?
A: To memoize a function so that it's not recreated on every render
Explanation: The 'useCallback' hook in React is used to memoize a function so that it's not recreated on every render, which can improve performance by preventing unnecessary re-renders.

Q: What would you do if you encountered a bug in your code that you couldn't reproduce?
A: Write a test case to reproduce the bug
Explanation: When encountering a bug that can't be reproduced, writing a test case to reproduce the bug is a good approach, as it allows you to isolate the issue and identify the root cause.

Q: How would you handle a situation where a stakeholder is requesting a feature that you believe will negatively impact the user experience?
A: Explain your concerns to the stakeholder and try to find a compromise
Explanation: When faced with a situation where a stakeholder is requesting a feature that may negatively impact the user experience, it's essential to explain your concerns and try to find a compromise that meets both the stakeholder's needs and the user's needs.

Q: What would you do if you were working on a team project and a team member was not pulling their weight?
A: Talk to the team lead about the issue
Explanation: When working on a team project and a team member is not pulling their weight, it's essential to talk to the team lead about the issue, as they can help mediate the situation and find a solution.

Q: What is the primary advantage of using a microservices architecture?
A: It allows for easier horizontal scaling
Explanation: The primary advantage of using a microservices architecture is that it allows for easier horizontal scaling, as each microservice can be scaled independently without affecting the entire system.

Q: What is the primary disadvantage of using a monolithic architecture?
A: It is difficult to scale
Explanation: The primary disadvantage of using a monolithic architecture is that it is difficult to scale, as the entire system must be scaled together, which can lead to performance issues and decreased system reliability.

Q: What is the primary advantage of using a caching layer in a system?
A: It improves system performance
Explanation: The primary advantage of using a caching layer in a system is that it improves system performance, as it reduces the number of requests made to the underlying system, which can lead to faster response times and improved user experience.

Problem: Write a function that takes a string as input and returns the string with all vowels removed. The function should handle both lowercase and uppercase vowels.

Solution:
```
def remove_vowels(input_string):
  vowels = 'aeiouAEIOU'
  return ''.join([char for char in input_string if char not in vowels])
```

Problem: Write a function that takes a list of integers as input and returns the sum of all the even numbers in the list. The function should handle empty lists and lists with non-integer values.

Solution:
```
def sum_even_numbers(numbers):
  return sum([num for num in numbers if isinstance(num, int) and num % 2 == 0])
```